CSE555 Final Project Report

EEG Eye State Dataset Analysis

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Abstract

This report analyzes the EEG Eye State dataset using statistical analysis, dimensionality reduction, clustering, and hypothesis testing. The dataset contains 14 EEG signal features and one binary class label [1] that shows whether the eyes are open or closed. The main goal is to understand the feature patterns, check outliers, compare class separation methods, and interpret the results with clear explanations. Boxplots and the IQR method showed that the data contains many extreme values, mostly in frontal and antero-frontal channels. After clipping, the data became easier to interpret. Correlation and Fisher Distance results showed that frontal and antero-frontal channels are more informative than other channels. PCA reduced the data dimension, but it did not separate the two classes clearly because PCA does not use class labels. LDA gives better class separation because it uses label information. The clustering results showed that the two eye-state classes still overlap, which means the problem is not naturally separated into two clean groups.

Keywords: EEG, eye state, outlier detection, Fisher Distance, PCA, LDA, clustering, t-test

1. Introduction

Electroencephalography (EEG) signals record brain activity from different electrode positions. The independent label is **eye_state**, where 0 means eye-open and 1 means eye-closed. The dataset contains 14 EEG channels: AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, and AF4.The general information about the dataset is shown in Table 1. [1]

Table 1. General dataset summary.

Property	Value
Number of samples	14,980
Number of numerical features	14
Target variable	eye_state
Class 0	Eye-open
Class 1	Eye-closed
Missing values	0
Main task	Binary classification

The analysis includes outlier detection, data cleaning, correlation analysis, z-score normalization, Fisher Distance, PCA, LDA, clustering, and hypothesis testing. These methods are used together because one feature alone is not enough to explain the two eye states clearly.

2. Boxplot Analysis and Outlier Detection

2.1 Boxplots Before Cleaning

The boxplots are important because they show the raw behavior of the dataset before any correction. In Figure 1, the boxes are very small compared with the full vertical range in some channels. This happens because a few very large or very small values force the axis to become wide. So, the normal values become visually compressed.

These extreme values should not be ignored directly, since EEG data can naturally contain sudden changes. At the same time, they should not dominate all later analysis. If the outliers stay as they are, methods such as correlation, PCA, clustering, and t-test can be affected by those extreme points. This is why I first checked the outliers before using more advanced methods.

The reason frontal and antero-frontal channels are more affected may be related to their position. These channels are closer to eye movement and frontal muscle activity. Eye blinks, small face movements, or electrode noise can create strong values in these electrodes. Because the dataset is about eye-open and eye-closed states, it is reasonable that these channels contain both useful information and more extreme values. [2]

Boxplots were first used to see the distribution of each EEG feature for the two eye states. Figure 1 shows that the two classes have similar distributions in many channels. This makes direct visual separation difficult.

The first important finding is that the dataset contains many extreme values. These outliers stretch the y-axis and make the normal part of the data look compressed. This is why some class differences are hard to see before cleaning. In EEG data, this kind of extreme value can happen because of eye blinks, small movements, electrode contact problems, recording noise, or sudden changes in brain signals.

The frontal and antero-frontal channels show the clearest difference between eye-open and eye-closed samples. For example, **antero_frontal_right_4**, **antero_frontal_left_3**, **frontal_right_4**, and **frontal_right_8** tend to have higher values for the eye-closed class. Temporal and occipital channels overlap more, so they are less useful for separating the classes by themselves.

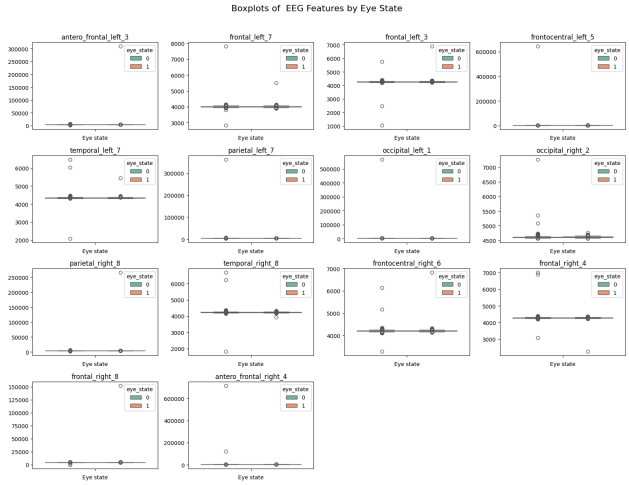


Figure 1. Boxplots of the original EEG features by eye state.

2.2 IQR Outlier Detection and Cleaning

I used IQR clipping instead of deleting samples because deleting too many rows could reduce the size of the dataset and change the balance between the two classes. Clipping keeps the sample in the dataset, but it limits the extreme value so it does not control the analysis too much.

The outlier counts in Figure 2 and Table 2 show that outliers are not only in one channel. They appear in every feature, but some channels have much more. This means the noise problem is a dataset-level issue, not only a single wrong column.

After clipping, the data is not made perfect. It is only made more stable for the next steps. The important part is that the class patterns that remain after clipping are more trustworthy, because they are less dependent on a small number of extreme samples.

Outliers were detected with the Interquartile Range (IQR) rule. Values below the lower boundary or above the upper boundary were treated as extreme values. After that, clipping was applied. This means that values smaller than the lower boundary were replaced by the lower boundary, and values larger than the upper boundary were replaced by the upper boundary. [3]

The outlier count for each channel is shown in Figure 2. The summary of the largest and smallest outlier counts is given in Table 2.

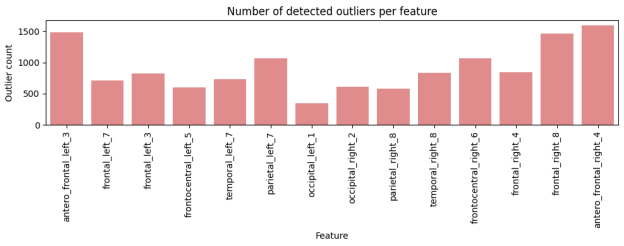


Figure 2. Number of detected outliers for each EEG feature.

Table 2. Highest and lowest detected outlier counts.

Feature	Outlier count	Comment
antero_frontal_right_4	1,592	Highest outlier count
antero_frontal_left_3	1,485	Very high outlier count
frontal_right_8	1,460	Very high outlier count
parietal_left_7	1,065	High outlier count
frontocentral_right_6	1,063	High outlier count
occipital_left_1	348	Lowest outlier count

The highest outlier counts are mostly in frontal or antero-frontal related channels. This supports the idea that these positions contain stronger activity changes and also more possible artifacts. After clipping, the same IQR rule no longer marked extreme outliers in the cleaned data.

2.3 Boxplots After Cleaning

After cleaning, Figure 3 makes the comparison between the eye states clearer. The boxes and medians can now be seen more easily. This helps to understand whether the two classes have a real distribution shift or whether the difference was only caused by extreme values.

The frontal and antero-frontal channels still show small differences after cleaning. This is important because it means those channels still carry useful class-related information. However, the overlap is still large, so these boxplots also show why a single feature is not enough for reliable classification.

Figure 3 shows the boxplots after IQR clipping. The distributions are much easier to read because the extreme values no longer dominate the scale. The class difference in frontal and antero-frontal channels still remains after cleaning, so the difference is not only caused by outliers.

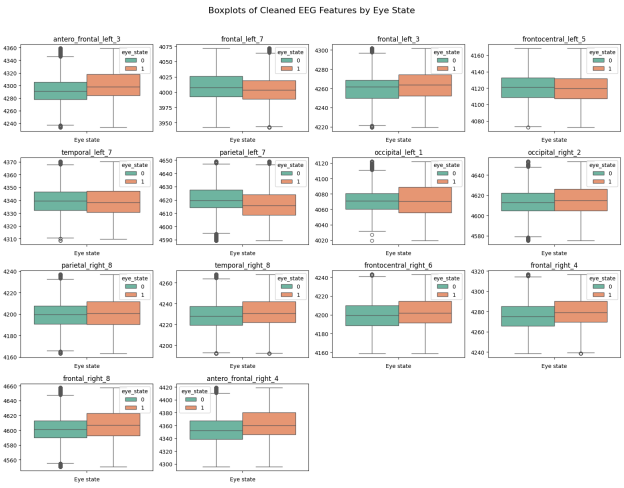


Figure 3. Boxplots of cleaned EEG features after IQR clipping.

Overall, the boxplot analysis shows that preprocessing is necessary before deeper analysis. Without cleaning, the extreme values can hide the useful information in the main distribution of the features.

3. Correlation Analysis of Features and Class Label

3.1 Correlation Among EEG Features

Pearson correlation was calculated on the cleaned dataset. The correlation values range from -1 to 1. A value close to 1 means that two features increase together, a value close to -1 means that one increases while the other decreases, and a value close to 0 means that there is little linear relationship. [4]

The heatmap in Figure 4 shows that many EEG features are not independent from each other. This is normal for EEG because nearby electrodes can measure related activity from the same brain region or from similar signal sources.

High correlation between features can be useful and problematic at the same time. It is useful because it shows a shared pattern in the EEG signals. But it can also mean that some features repeat similar information. This is one reason why dimensionality reduction methods such as PCA are useful later in the report.

The absence of strong negative correlations also gives information about the dataset. Most channels tend to change in a similar direction instead of opposite directions. This supports the idea that eye-state information is spread across several channels rather than appearing as one isolated signal.

Figure 4 shows the correlation heatmap. Many nearby channels are strongly positively correlated. For example, antero_frontal_left_3 and antero_frontal_right_4 have a correlation of 0.93. This is expected because nearby electrodes can record similar brain activity.

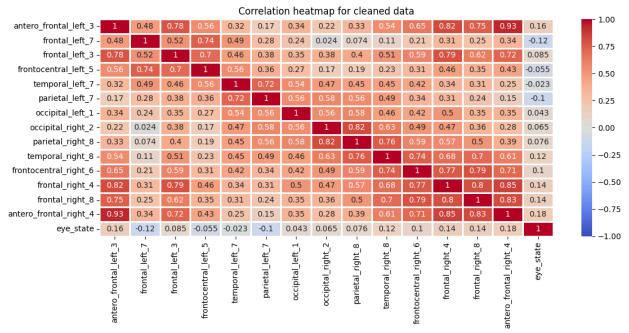


Figure 4. Correlation matrix of the cleaned EEG features and eye-state label.

Other strong relationships include frontal_right_4 with antero_frontal_right_4 (0.85), frontal_right_8 with antero_frontal_right_4 (0.83), occipital_right_2 with parietal_right_8 (0.82), and frontal_right_4 with frontal_right_8 (0.80). There were no strong negative correlations, so most EEG features tend to move in the same general direction.

3.2 Correlation with Eye State

The label correlations in Figure 5 and Table 3 are not high. The highest absolute correlation is still below 0.20. This means there is no single EEG feature that has a strong linear relationship with eye_state.

This weak correlation does not mean the dataset is useless. It means the relationship is subtle. A feature can still be helpful when it is combined with other features. For example, several frontal features have small positive correlations, and together they can provide more information than each feature alone.

The negative correlations also need to be read carefully. A negative value means the feature tends to decrease when moving from class 0 to class 1. It does not mean the feature is bad. It only shows the direction of the relationship with the eye-state label.

Figure 5 shows the correlation between each EEG feature and the eye_state label. The strongest positive values belong to antero_frontal_right_4, antero_frontal_left_3, frontal_right_4, and

frontal_right_8. These features tend to be higher when the eyes are closed.

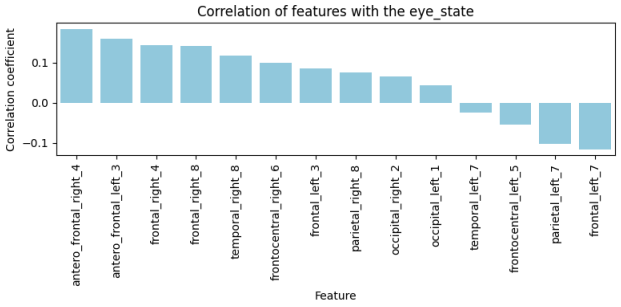


Figure 5. Correlation coefficients between EEG features and the eye-state class label.

Table 3. Important positive and negative correlations with the eye-state label.

Feature	Correlation	Direction
antero_frontal_right_4	0.184	Positive
antero_frontal_left_3	0.160	Positive
frontal_right_4	0.144	Positive
frontal_right_8	0.141	Positive
frontal_left_7	-0.115	Negative
parietal_left_7	-0.103	Negative
frontocentral_left_5	-0.055	Negative

The correlation values with the label are still weak because none of them is above 0.20 in absolute value. This means one EEG channel is not enough to classify eye state reliably. A combination of features is needed.

4. Z-Score Normalization and Fisher Distance Analysis

4.1 Z-Score Normalization

Normalization was necessary because all EEG channels do not have the same scale and variation. If one feature has a larger range than the others, it can dominate distance-based or variance-based methods. Therefore, z-score normalization was applied to put all features on a comparable scale, as shown in Equation (1). After z-score normalization, each feature has a mean close to 0 and a standard deviation close to 1. This process does not remove the original pattern of the data. It only changes the scale of the features, so the methods can compare them more fairly. [5]

$$(1) \quad z = (x - \mu) / \sigma$$

Equation (1). Z-score normalization, where x is the original value, μ is the feature mean, and σ is the standard deviation.

4.2 Fisher Distance

Fisher Distance was used because it gives a direct measure of how much one feature separates the two classes. It compares the distance between class means with the variation inside each class. The idea is shown in Equation (2). [6]

A high Fisher Distance means the two classes have means that are more separated compared with their internal spread. A low value means the two classes are highly mixed for that feature. This is why Fisher Distance is more useful than only looking at the mean difference.

The ranking in Figure 6 and Table 4 agrees with the correlation result. Antero-frontal and frontal channels are again the most informative.

Still, the Fisher Distance values are not very large. This means the features are useful, but not strongly separated.

Fisher Distance measures how well one feature separates two classes. A larger value means the class means are more different compared with the variation inside the classes. The formula used for the two class case is shown in Equation (2).

$$(2) \quad FD = (\mu_0 - \mu_1)^2 / (\sigma_0^2 + \sigma_1^2)$$

Equation (2). Fisher Distance for two classes.

The Fisher Distance values are shown in Figure 6. The most informative features are antero_frontal_right_4 (0.071), antero_frontal_left_3 (0.053), frontal_right_4 (0.043), and frontal_right_8 (0.041).

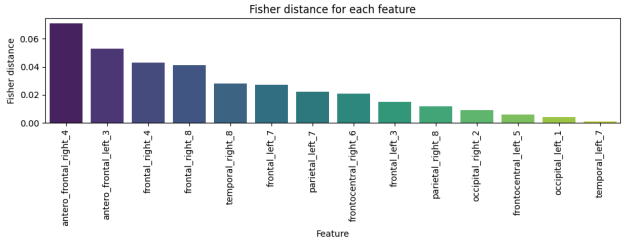


Figure 6. Fisher Distance values calculated for all normalized EEG features.

Table 4. Fisher Distance ranking for selected normalized EEG features.

Feature	Fisher Distance	Interpretation
antero_frontal_right_4	0.071	Highest separation
antero_frontal_left_3	0.053	Strong useful information
frontal_right_4	0.043	Useful information
frontal_right_8	0.041	Useful information
temporal_right_8	0.028	Moderate information
frontal_left_7	0.027	Moderate information
parietal_left_7	0.022	Limited but useful information
frontocentral_right_6	0.021	Limited but useful information
frontocentral_left_5	0.006	Very low separation
occipital_left_1	0.004	Very low separation
temporal_left_7	0.001	Lowest separation

The Fisher Distance result agrees with the correlation analysis. The same frontal and antero-frontal channels appear near the top in both methods. This makes these channels important for later analysis, but the values are still not high enough to show perfect separation.

5. PCA Transformation and Fisher Distance Analysis

5.1 PCA Purpose

PCA was used to reduce the number of features and to see the main directions of variation in the EEG data. Since many EEG channels are correlated, PCA can summarize repeated information with fewer components. The PCA projection idea is shown in Equation (3). [7]

It is important to remember that PCA does not know the eye_state labels. It only looks at variance. Because of this, the component that explains the most variance is not always the component that separates eye-open and eye-closed samples best.

Principal Component Analysis (PCA) transforms correlated features into new uncorrelated features called principal components. PCA is useful because many EEG channels are related to each other. The basic projection idea is shown in Equation (3), where the normalized data is projected onto PCA directions.

$$(3) \quad Z = XW$$

Equation (3). PCA projection, where X is normalized data, W contains PCA component directions, and Z is the projected data.

Since, PCA does not use the eye_state labels and only tries to preserve the largest variance in the dataset for this reason, a component with high variance is not always the best component for separating classes.

5.2 PCA Variance Analysis

The variance result in Figure 7 shows that the first components carry most of the information. This is expected because several EEG channels are strongly related to each other. PCA can combine this shared information into a smaller number of components.

However, high explained variance mostly means that the component describes a strong pattern in the whole dataset. It does not automatically mean that the pattern is useful for the class label. A high variance component can capture general EEG activity, sensor variation, or common movement in channels, not only eye-state difference.

Figure 7 shows the cumulative explained variance by the PCA components. The first components capture most of the total variance. This means the EEG channels share common information and the data can be represented with fewer components.

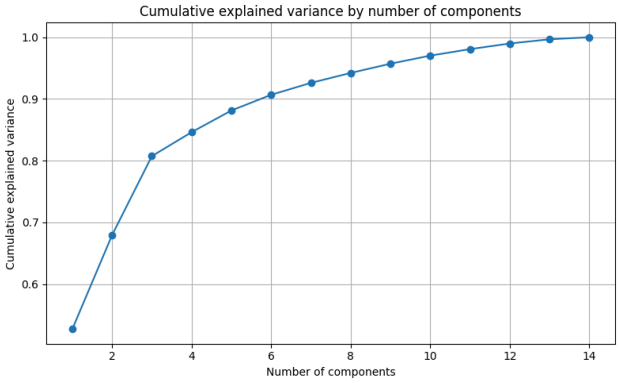


Figure 7. Cumulative explained variance by number of PCA components.

5.3 Fisher Distance of PCA Components

The PCA Fisher Distance result in Figure 8 and Table 5 is one of the most important parts of the report. PC1 has the highest eigenvalue, but PC3 has the highest Fisher Distance. This shows clearly that variance and class separation are not the same thing.

PC3 may capture a smaller but more class related direction. This means it does not explain the biggest change in the whole dataset, but it explains a change that is more useful for eye-open and eye-closed separation. This is why PCA components should not be selected only by eigenvalue when the purpose is classification.

PCA can still be useful because PC3 reached a Fisher Distance of 0.093, which is higher than the best single original feature. This means PCA can combine information from several channels and create a component that is more informative than one feature alone.

After PCA, Fisher Distance was calculated for each principal component. The relationship between eigenvalues and Fisher Distance is shown in Figure 8. The figure shows that large variance and class separation are not always the same thing.

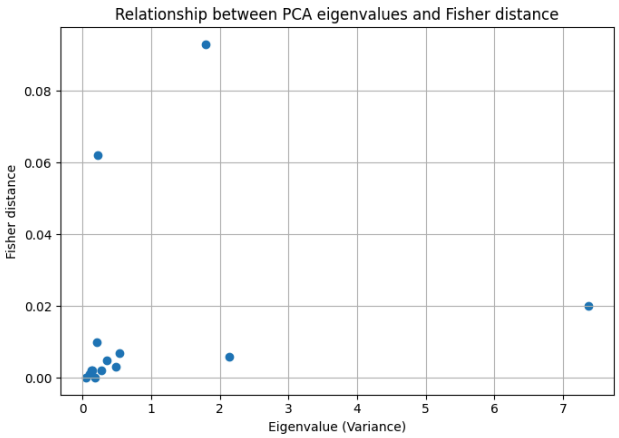


Figure 8. Relationship between PCA eigenvalues and Fisher Distance.

Table 5. Selected PCA component results

Component	Eigenvalue	Explained variance ratio	Fisher Distance
PC1	7.376	0.527	0.020
PC2	2.134	0.152	0.006
PC3	1.792	0.128	0.093
PC8	0.226	0.016	0.062
PC14	0.046	0.003	0.000

PC1 explains the largest variance, but it does not give the highest Fisher Distance. PC3 gives the highest Fisher Distance among the shown PCA components. This happens because PCA is unsupervised. It searches for directions with high variance, not directly for directions that separate eye-open and eye-closed states.

Compared with the original features, PCA can combine information from several EEG channels. The strongest original feature had a Fisher Distance of 0.071, while PC3 reached 0.093. This shows that class-related information is distributed across multiple channels.

6. Visualization of PCA Projections

6.1 Projection onto the First Two Components

In Figure 9, the first two PCA components show some structure, but the two colors still overlap. This happens because PCA is a linear method and it only focuses on variance. The eye-state classes are not separated by one simple direction in the first two components.

The shape of the scatter plot also suggests that the EEG samples change continuously. Instead of two clear clouds, the data forms mixed regions. This supports the earlier finding that eye-state information is subtle and distributed across many features.

Figure 9 shows the projection onto PC1 and PC2. These two components explain a large part of the variance. The plot shows partial separation between the eye-open and eye-closed classes, but the classes still overlap strongly.

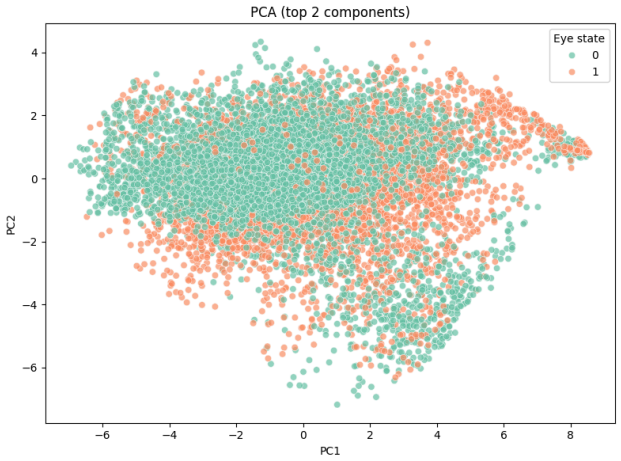


Figure 9. Projection of the EEG dataset onto the first two principal components.

The overlap is expected because PCA only uses variance, not labels. Some of the large variance in EEG data can come from general signal changes, noise, or individual sample differences, not only from eye state.

6.2 Projection onto the Least Important Components

In Figure 10, the least important PCA components show almost no class separation. These components have very small eigenvalues, so they describe weak variations left after the main patterns are removed.

This does not mean the last components are always useless in every problem, but in this dataset they mostly represent small changes, noise, or information that is not strongly related to eye_state. Therefore, they are not very helpful for visual separation.

The comparison between Figure 9 and Figure 10 shows why component selection matters. The first components are better for representing the global structure, while the last components are much less useful for interpretation.

Figure 10 shows the projection onto the two least important components. The classes are almost completely mixed. These components have small eigenvalues, so they contain weak variations, noise, or information that is not useful for classification.

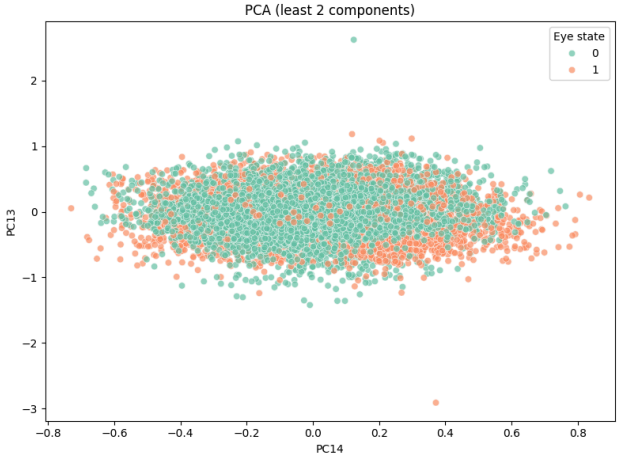


Figure 10. Projection of the EEG dataset onto the two least important principal components.

The comparison between Figure 9 and Figure 10 confirms that the first components are more useful for understanding the data structure. Still, PCA alone cannot create a clean boundary between the two classes.

7. LDA-Based Visualization and Comparison with PCA

7.1 LDA Method

LDA is different from PCA because it uses the class labels while creating the projection. Its objective is to find the direction where the two class means are far from each other and where each class is as compact as possible. This idea is written in Equation (4). [6]

Because there are only two classes in this dataset, LDA can create only one main discriminant direction. Means LDA tries to find the single best line for separating eye-open and eye-closed samples.

Linear Discriminant Analysis (LDA) is a supervised dimensionality reduction method. Unlike PCA, it uses the class labels. Its goal is to maximize the difference between classes and minimize variation inside each class. The LDA Fisher criterion is shown in Equation (4).

$$(4) \quad J(w) = (w^T S_B w) / (w^T S_W w)$$

Equation (4). LDA objective, where S_B is between-class scatter and S_W is within-class scatter.

Because this dataset has only two classes, LDA can produce one discriminant component. A small vertical jitter was added only for visualization so that the one dimensional projection can be seen more clearly.

7.2 LDA Projection and Comparison

The LDA projection in Figure 11 is more separated than the PCA projection. This is expected because LDA was designed for class separation, while PCA was designed for variance preservation.

The numerical result in Table 6 also supports this interpretation. The best Fisher Distance increases from 0.093000 in PCA to 0.230000 in LDA, and the silhouette score also increases. These values show that LDA gives a more discriminative representation.

However, the LDA plot is still not perfectly separated. This is important to mention honestly. Even when labels are used, the EEG patterns overlap because the signal is noisy and the difference between eye-open and eye-closed states is not strong in every sample.

Figure 11 shows the LDA projection. Compared with the PCA plot, LDA gives a more organized separation between the two classes. Some overlap still remains, but the class direction is clearer.

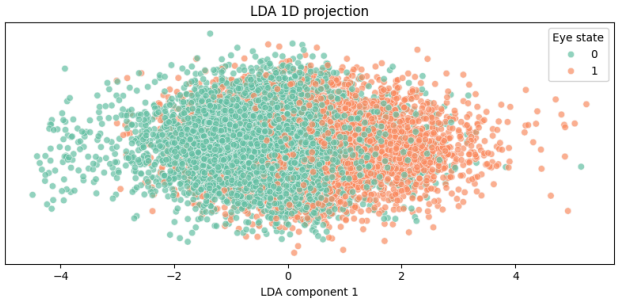


Figure 11. One-dimensional representation of the EEG dataset obtained using Linear Discriminant Analysis.

Table 6. PCA vs LDA performance comparison

Metric	PCA (unsupervised)	LDA (supervised)
Best Fisher Distance	0.093000	0.230000
Silhouette score	0.014459	0.078509

The numerical comparison in Table 6 supports the visual result. LDA has a higher best Fisher Distance and a higher silhouette score than PCA. This means LDA is more useful when the main goal is class separation.

The remaining overlap is also important. It shows that eye-open and eye-closed EEG samples are not perfectly separable with a simple linear projection. It shows that the data is nonlinear.

8. Clustering Analysis Using K-Means, DBSCAN, t-SNE, and SOM

8.1 K-Means and DBSCAN on PCA Space

K-Means and DBSCAN were used to test whether the data naturally forms groups without looking at the labels. This is different from classification. In clustering, the algorithm does not know which sample is eye-open or eye-closed. The labels are only used later to evaluate the clusters. [8], [9]

In Figure 12, K-Means finds two large geometric regions, but these regions do not match the two eye-state labels clearly. This means K-Means is separating the space based on distance, not necessarily based on the real biological class.

In Figure 13, DBSCAN finds dense areas and some smaller regions, but it also cannot create clean eye-state groups. This is because the data does not have two separated density areas. The eye-open and eye-closed samples are mixed inside the same continuous space.

Clustering was used to check whether the data naturally forms groups without using labels during training. For K-Means and DBSCAN, the first two PCA components were used. The true labels were only shown in the interpretation stage.

Figure 12 shows K-Means clustering with two clusters. K-Means finds two large regions, but both regions contain mixed eye states. This happens because K-Means depends on distance to cluster centers and assumes simple cluster shapes.



Figure 12. K-Means clustering performed on the first two principal components.

Figure 13 shows DBSCAN clustering. DBSCAN groups dense regions and can mark noise points. It detects some density structure, but the two eye-state classes still overlap a lot. The result is strongly affected by the epsilon and minimum-samples parameters.

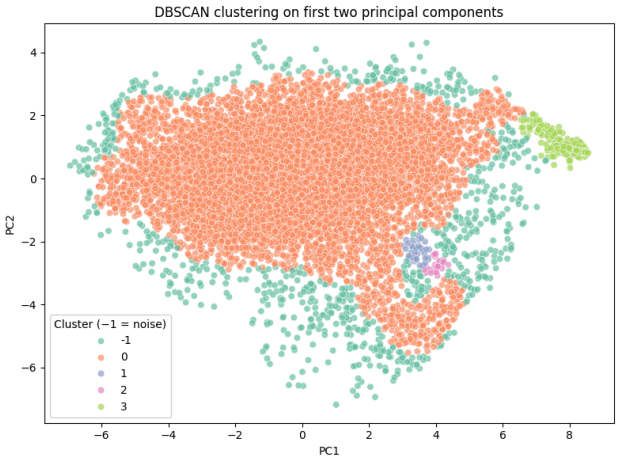


Figure 13. DBSCAN clustering performed on the first two principal components.

8.2 t-SNE and SOM Visualization

t-SNE in Figure 14 gives a more detailed local view. Some same-label samples become closer to each other, but the labels still do not form two completely separate areas. This means nonlinear visualization helps, but it does not solve the overlap completely.

The SOM dominance map in Figure 15 is shows which nodes are more related to eye-open or eye-closed samples. The sample count map in Figure 16 is also showing a node with a strong class value may not be important if it has only a few samples.

Together, t-SNE and SOM show that there is some hidden structure in the dataset. But they also confirm that the labels are mixed. So, these visualizations are useful for understanding the data, but they are not enough alone for classification.

t-SNE was applied to the original normalized features. Unlike PCA, t-SNE is nonlinear and focuses on local neighborhood relationships. As shown in Figure 14, it reveals more local structure than PCA, but the two labels still overlap. [10]



Figure 14. Two-dimensional t-SNE representation of the original normalized EEG dataset.

A Self-Organizing Map (SOM) was also used. Figure 15 shows the node dominance map based on the mean eye_state value, and Figure 16 shows how many samples are assigned to each SOM node. Together, these maps show both class tendency and data density. [11]

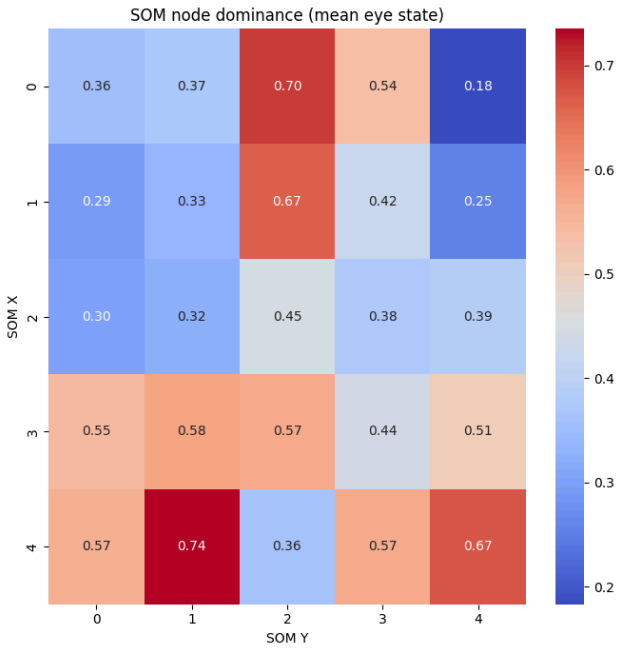


Figure 15. SOM node dominance map based on mean eye_state values.

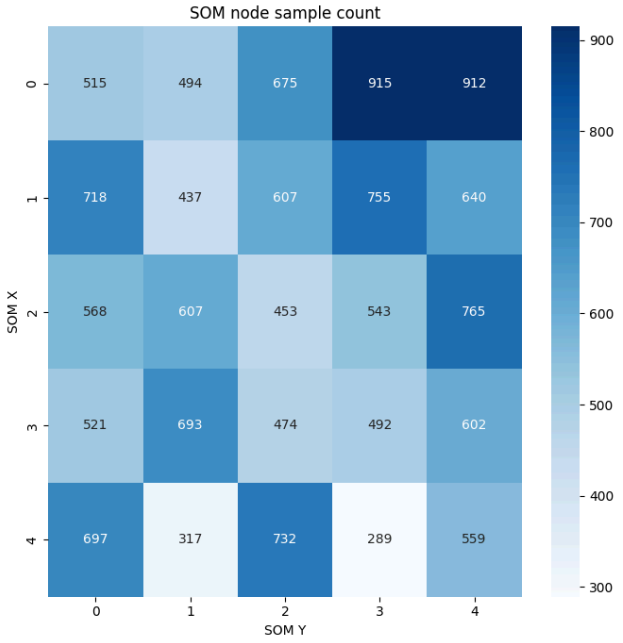


Figure 16. SOM node sample count map.

8.3 Clustering Evaluation

The clustering metrics in Table 7 make the visual interpretation clearer. The K-Means silhouette score is not too low, so it did find a geometric separation. But the ARI and NMI values are very low, which means the clusters do not agree with the actual labels.

This difference between silhouette and ARI/NMI is important. A good silhouette score only means clusters are compact and separated in the chosen feature space. It does not guarantee that those clusters represent eye-open and eye-closed classes. [12]

For DBSCAN, the ARI and NMI are also very low. This supports the same conclusion: the EEG Eye State dataset has some structure, but the true class labels are not naturally split into clean unsupervised clusters.

Table 7. Quantitative clustering evaluation results.

Method	ARI	NMI	Silhouette score
K-Means	0.0172	0.0086	0.4511
DBSCAN non-noise samples	0.0026	0.0014	0.3377

The K-Means silhouette score is 0.4511, which means the algorithm found a visible geometric structure in PCA space. However, the ARI and NMI values are very low. This means the K-Means clusters do not match the real eye-state labels very well.

DBSCAN also has very low ARI and NMI values. This shows that it found density-based structure, but that structure is not the same as the true eye-open and eye-closed classes. Overall, the clustering results confirm that the dataset contains class-related information, but not as two clean natural clusters.

9. Hypothesis Testing Using the t-Test

9.1 Purpose and Selected Feature

For the hypothesis test I selected **antero_frontal_left_3** because it appeared many times as an informative feature. It had a positive correlation with eye_state, a relatively high Fisher Distance, and it is located in the antero-frontal region, which was important in earlier sections.

This does not mean this feature is the only important feature. It was selected as an example feature for statistical testing. The main idea is to test whether one of the informative features really has a measurable mean difference between the two classes.

Hypothesis testing was used to check whether the mean value of one EEG feature is significantly different between the eye-open and eye-closed classes. The independent two-sample test was used because the eye-open samples and eye-closed samples are two different groups. Since the two groups may not have exactly the same variance, the full-data interpretation uses Welch's version of the t-test. The test statistic is shown in Equation (5). [13]

$$(5) \quad t = (\bar{x}_0 - \bar{x}_1) / \mathrm{sqrt}(s_0^2/n_0 + s_1^2/n_1)$$

Equation (5). Welch independent t-test statistic for comparing the two class means.

9.2 Hypotheses

The null hypothesis, H0, assumes that there is no mean difference between eye-open and eye-closed samples for this feature. The alternative hypothesis, H1, assumes that there is a difference. Because the direction was not forced before the test, the test was written as a two-sided test.

A two sided test is more careful here because it checks for any difference, not only whether one class is higher than the other. After the test, the sign of the t-statistic and Cohen's d can be used to understand the direction of the difference.

The p-value tells whether the observed difference is statistically significant. But it does not tell how large or useful the difference is. For that reason, Cohen's d was also included through Equation (6).

H0: μ_open = μ_closed

H1: μ_open ≠ μ_closed

The significance level was set to $\alpha = 0.05$. If the p-value is smaller than 0.05, H0 is rejected. If the p-value is larger than 0.05, H0 is not rejected.

Cohen's d was also used to interpret the effect size, because a very small p-value does not always mean a large practical difference. The effect size formula is shown in Equation (6).

$$(6) \quad d = (\bar{x}_0 - \bar{x}_1) / s_pooled$$

Equation (6). Cohen's d effect size for the difference between the two class means.

9.3 t-Test Results and Interpretation

The results in Table 8 should be read in two parts. The first two rows are random sample experiments, and the last row is the full cleaned dataset. The full cleaned dataset is the most stable result because it uses all available data instead of a small random subset.

The 36-sample test rejects H0, but the 64-sample test does not reject H0 at the 0.05 level. This does not mean the analysis is wrong. It shows that small random samples can change the result when the class difference is not very large. This can happen because random sampling may select slightly different parts of the distribution. When the real class difference is small, some samples can show a clearer difference, while others can look closer together.

The full cleaned dataset gives the most stable result. The full-data test has t = -19.8745 and p = 9.58e-87, so H0 is rejected. This means that antero_frontal_left_3 has a statistically significant mean difference between eye-open and eye-closed states.

The negative t-statistic and negative Cohen's d show the direction of the difference. In this output, the mean value for class 0 is lower than the mean value for class 1. Since class 0 is eye-open and class 1 is eye-closed, antero_frontal_left_3 tends to be higher when the eyes are closed.

The most important point is that the p-value is extremely small partly because the full dataset is large. Cohen's d = -0.3255 is not a large effect. Therefore, the feature is statistically useful, but it cannot separate the two classes by itself. This supports the earlier correlation, Fisher Distance, PCA, and LDA findings: eye-state information is spread across multiple EEG channels.

Table 8. Independent t-test results for antero_frontal_left_3.

Experiment	Data size	t-statistic	p-value	Cohen's d	Decision
Random sample 1	36 per class	-3.302	0.0015	-0.778	Reject H0
Random sample 2	64 per class	-1.906	0.0590	-0.337	Fail to reject H0
Full cleaned dataset	All cleaned samples	-19.8745	9.58e-87	-0.3255	Reject H0

In Table 8, the first random sample shows a significant result and a stronger effect. The second random sample is close to the 0.05 boundary but is not significant. The full cleaned dataset is the most reliable result because it uses all available cleaned samples.

10. Conclusion

The project analyzed the EEG Eye State dataset with several statistical and machine learning methods. The first part showed that the dataset contains many outliers, especially in frontal and antero-frontal channels. IQR clipping made the distributions more compact and easier to interpret.

Correlation and Fisher Distance results showed that antero-frontal and frontal features are the most informative channels. However, the correlations with the label were weak, so no single channel can classify the two eye states perfectly.

PCA was useful for reducing dimensionality, but it did not fully separate the classes because it does not use label information. The PCA results also showed that high variance does not always mean high class separation. PC3 had the highest PCA Fisher Distance even though PC1 had the largest eigenvalue.

LDA gave better class separation than PCA because it uses the class labels. The best Fisher Distance increased from 0.093000 for PCA to 0.230000 for LDA, and the silhouette score increased from 0.014459 to 0.078509. Still, some overlap remained because EEG signals are noisy and the class differences are subtle.

The clustering results also supported this conclusion. K-Means and DBSCAN found some geometric and density-based structures, but their ARI and NMI values were very low. This means the unsupervised clusters did not strongly match the real eye-state labels.

Finally, the hypothesis testing section showed that antero_frontal_left_3 has a statistically significant mean difference in the full cleaned dataset. However, the effect size was small to moderate, so this feature is useful but not enough alone. Overall, the dataset contains meaningful information about eye state, but the two classes remain overlapping.

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